

CoRoT-2b

R-1101

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_160723 · created 2026-07-30T19:17:01+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457592.802 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.004 +/- 0.064
Transit depth (Rp/Rs)^2	0.0015 +/- 0.049 [%]
Orbital Inclination (inc)	85.19 +/- 2.08
Airmass coefficient 1 (a1)	0.78 +/- 0.48
Airmass coefficient 2 (a2)	0.27 +/- 0.94
Scatter in the residuals of the lightcurve fit is	65.66 %
Best Comparison Star	None
Optimal Aperture	1.22
Optimal Annulus	3.0
Transit Duration (day)	0.068 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.004 ± 0.064 vs 0.1667 (deviation 2.54σ)
Duration fit vs published	0.068 d vs 0.095 d (deviation 1.5σ)
Mid-time O–C	-5.3 min
Depth SNR	0.1
Residual scatter	65.66 %

Light curve

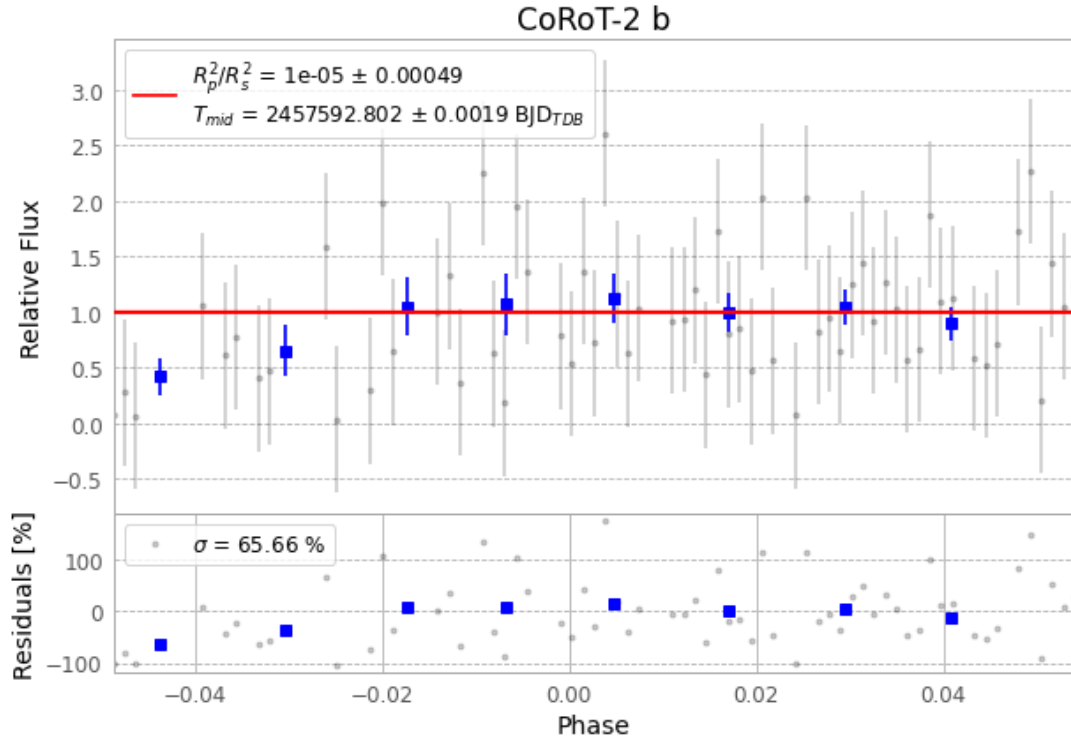


Figure 1 — FinalLightCurve_CoRoT-2 b_2016-07-22.png (SHA-256 0c05bd67d01b1121...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [221, 209], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 47e7e3f36f9b3a5a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

87 FITS frames (57 MB), CoRoT-2160723050312.FITS → CoRoT-2160723092112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-160723.log (731cd087666f...), FinalLightCurve_CoRoT-2 b_2016-07-22.png (0c05bd67d01b...), FinalLightCurve_CoRoT-2 b_2016-07-22.csv (b72ebc416eeb...)

Compute resources

Wall time 210.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 19:17Z.